Transport-scheme comparison — chain vs. SC vs. octree, with and without delta-M, ranked against a fine reference

Comparison of the three transport configurations of the `rt3d` solver on the MANUAL §11 worked example, each run with and without delta-M (Wiscombe) scaling, **plus a fine-octree reference solution and grid-refinement runs that rank every configuration by accuracy** (plan: `plan/compare-transport-plan.md` + `plan/transport-ranking-plan.md`, incl. its addendum §7); all figures and CSVs live in `plot/compare_transport/`. The two runs on the fine reference octree use **642 ordinates** (icosahedral refinement 3) instead of the 42-ordinate development grid, so the reference is finer in *angle* as well as in space and the ranking errors of the 42-ordinate runs include their angular error (comparisons *among* the 42-ordinate runs still cancel it). Measured 2026-10-15 on macOS/arm64 (Apple M4 Pro, 64 GB) with the `pyarts` interpreter; physics numbers are machine-independent, wall times are not (the previous version of this report was measured on Windows — wall times differ by ~2–4× between the two machines; Ng-accelerated iteration counts can also shift by a few iterations with BLAS rounding, e.g. SC δ-off: 44 → 51). The wall times here predate the numba fast paths of `rt3d/jit.py` (landed after this run started); reruns with them will be faster.

Reproduce with:

```
~/miniforge3/envs/pyarts/bin/python scripts/compare_transport.py
```

(The 642-ordinate reference runs need ~22 GB RAM for their octree march tables — ~34 MB per ordinate on the 50 380-leaf grid. 3D HTML scenes need `trame-pyvista`; installed in the macOS `pyarts` env on 2026-10-13, so all scenes and `vol_j.png` are current. Per-run solutions are cached in `plot/compare_transport/_solve_cache/` (gitignored) — reruns skip the solves; the cache records the ordinate count and stale entries are re-solved.)

---

1. Purpose and run matrix

The solver currently offers three transport algorithms and an optional delta-M split of the scattering operator. This study answers four questions on one physically non-trivial case (strongly forward-scattering cloud, solar beam, reflecting ground):

1. **chain vs. SC** — how large is the pure *transport-algorithm* error when both run on *identical cells*?
2. **unigrid vs. octree** — the grid-refinement error (validation anchor 5).
3. **delta-M on/off** — does it accelerate convergence, and does the converged solution stay the same (the equivalence claim of `plan/delta-m-plan.md`)?
4. **which run is best?** — pairwise differences alone cannot answer this (a gap between two approximations says nothing about which is closer to the exact solution), so the study adds a fine reference solution and ranks every run by its error against it (section 5).

Six comparison runs plus five ranking runs (N_dir column: icosahedral refinement 1 = 42 ordinates for all runs except the two on the fine reference octree, which use refinement 3 = 642 ordinates):

run key	grid	transport	delta-M	N_dir	role
`uni_chain_dM0`	unigrid 24³	lattice-stencil chains	off	42	comparison
`uni_chain_dM1`	unigrid 24³	lattice-stencil chains	M = 1	42	comparison
`uni_sc_dM0`	unigrid 24³	single-cell short characteristics	off	42	comparison
`uni_sc_dM1`	unigrid 24³	single-cell short characteristics	M = 1	42	comparison

run key	grid	transport	delta-M	N_dir	role
`octree_dM0`	octree (25 264 leaves)	exact characteristic marches	off	42	comparison
`octree_dM1`	octree (25 264 leaves)	exact characteristic marches	M = 1	42	comparison
`uni_chain48_dM1`	unigrid 48^3 (h = 0.208 m)	chains	M = 1	42	chain refinement
`uni_sc48_dM1`	unigrid 48^3	short characteristics	M = 1	42	SC refinement
`octree_d4_dM1`	octree τ_max = 1.0, depth 4 (34 476 leaves)	exact marches	M = 1	42	octree refinement
`octree_ref_dM1`	octree τ_max = 0.5, depth 5 (50 380 leaves)	exact marches	M = 1	**642**	fine-grid δ-M check
`octree_ref_dM0`	same octree as `octree_ref_dM1`	exact marches	off	**642**	**the reference**

---

2. The setups

2.1 Common physical case (MANUAL §11, unchanged)

Gaussian cloud + Lambertian ground + solar beam on the domain x, y ∈ [−5, 5], z ∈ [0, 10] m, periodic in x and y, space at the top, ground at the bottom. Units: lengths m, κ in m^{-1}, radiometric quantities normalised to the incident beam flux $F_0 = 1$.

- **Extinction** κ_e = C·gauss(σ = 1 m, μ = (0, 0, 4) m), C scaled so the maximum vertical column optical depth is τ = 10 (peak κ_e ≈ 3.9 m^{-1}, mean free path ≈ 0.26 m).
- **Scattering** κ_s = ω_local·κ_e with ω_local = gauss/max(gauss), capped at ω ≤ 0.9 (`omega_cap=0.9`) so plain lambda iteration stays tractable.
- **Asymmetry** g_local = gauss/(1.05·max) → peak g ≈ 0.952 in the cloud core (HG phase function) — the showcase regime for delta-M.
- **Thermal emission** B = 0 (pure scattering/beam problem).
- **Solar beam**: direct collimated beam, 30° from zenith, propagating in the x–z plane: ω_sun = [sin 30°, 0, −cos 30°], $F_0 = 1$.
- **Boundary conditions**: isotropic background I = 1 on the top face; Lambertian ground, albedo 0.3, emissivity 0 (no thermal emission).

2.2 Direction grids and iteration scheme

- Icosahedral direction grids, solid-angle weights from spherical-triangle areas: refinement 1 → **42 ordinates** (development grid, all comparison and refinement runs, Σw − 4π = −7.1e−15) and refinement 3 → **642 ordinates** for the two runs on the fine reference octree (Σw − 4π = +7.1e−15).
- Lambda iteration with **1D Ng acceleration**, convergence on the relative source change, tol = 1e−6, cap 60 iterations.
- The 42-ordinate *angular* error cancels in all comparisons *among* the 42-ordinate runs (the six base pairs, the unigrid refinement pairs), but **not** against the 642-ordinate reference: the ranking errors of the 42-ordinate runs include their angular error, and the reference's own angular error is ~15× better resolved (section 5.2).

2.3 Grids

- **Unigrid 24³**: h = 0.4167 m, 13 824 cells; **unigrid 48³**: h = 0.2083 m, 110 592 cells.
- **Octree** base cell h_0 = 1.25 m, refinement driven by the *physical* κ_e, 2:1 balanced:
  - depth 3 (τ_max = 2.0): **25 264 leaves** (L0: 316, L1: 788, L2: 3 680, L3: 20 480), cell size 1.25 m in the vacuum tails down to 0.156 m;
  - depth 4 (τ_max = 1.0): **34 476 leaves** (finest cells still 0.156 m — the extra refinement acts on the intermediate levels);
  - depth 5 (τ_max = 0.5): **50 380 leaves**, finest cells **0.078 m** (τ_cell ≤ 0.5 in the cloud core). κ_e-driven refinement *saturates* here: τ_max < 0.5 or depth > 5 adds no leaves, so this is the finest discretization the solver builds on this case — hence the reference.

2.4 Transport algorithms

- **`chain` (unigrid)** — lattice-stencil chain sweeps (`rt3d/characteristics.py`, DOIT-style): per ordinate one integer stencil s, characteristics run center-to-center across the whole grid in multi-cell chains. Exact for piecewise-constant sources along the chain; carries the known *grazing-ordinate stencil error* on coarse grids (MANUAL §6, anchor 5).
- **`sc` (unigrid)** — single-cell short characteristics (`rt3d/short_characteristics.py`, `plan/plan-sc-unigrid.md`): gap segment + target segment of equal length, bilinear 4-point face interpolation, wrap-around passes on periodic entry axes. Local and gather-only; more diffusive on coarse grids (path error ~ linear in cell size).
- **`Octree`** — exact characteristic marches (`transport.precompute_octree_march` + `sweep_direction_octre`) one backward ray march per leaf per ordinate along the *true* ordinate direction, no lattice-stencil approximation. Short characteristics are *not available* on the octree (the solver raises `ValueError` for `transport="sc"` there).

2.5 delta-M scaling (`delta_M=1`, delta-Eddington)

Wiscombe split of the HG phase function, defined on the *row-normalized* discrete operator (MANUAL §9, `plan/delta-m-plan.md`):

```
Phi_delta = Phi_HG - (1/4pi) * sum_{n<1} (2n+1) g^n P_n(Omega.Omega')
kappa_e_delta = kappa_e - kappa_s * g^M        (diffuse transport only)
```

with the explicit `-omega·g^M·I/(1 - omega·g^M)` term in the source. Only the scattering part of the extinction is scaled; the direct beam keeps its physical attenuation, and octree refinement stays driven by the physical κ_e. delta-M changes the *iteration* (fewer iterations at g ≈ 0.95), not the converged discrete solution — the on/off comparison below verifies this.

2.6 The reference solution

`octree_ref_dM0`: the depth-5 octree (h_min = 0.078 m ≈ 3.3 cells per mean free path in the cloud core) at **642 ordinates**, exact marches, **delta-M off** so it carries no O(τ_cell) quadrature gap of the explicit −ω g^M I term. It is the finest space-angle discretization available on this case; its own remaining error is bounded by the depth-3 → depth-4 → depth-5 *space* convergence sequence and by the 42 → 642 *angle* step (section 5.2: the 42-ordinate reference scalars differ from the 642-ordinate ones by ~1.6% in ⟨J⟩ and ~1% in the TOA flux — the reference's remaining angular error is correspondingly smaller), and by the fine-grid δ-on↔δ-off check (0.19% max, section 4.1).

---

3. Diagnostics

All cell fields are sampled onto one **common 64³ lattice** at the baseline octree cell size (h = 0.15625 m) — exact pointwise sampling for piecewise-constant fields, the finer reference cells are sampled, not averaged — so every pair of runs is directly pointwise comparable.

- **Mean intensity** $J = (1/4\pi)\int I\, d\Omega$.
- **Angular variance** $\sigma^2(r) = (1/4\pi)\int (I - J)^2\, d\Omega$ (`rt3d.diagnostics.angular_variance`) — zero for an isotropic field, measures the angular anisotropy of the radiation field.
- **Upward radiance at TOA**: the upward flux map $F^+(x, y) = \int_{\mu>0} I\, \mu\, d\Omega$ at the top layer, and the domain-mean upward radiance per upward ordinate (21 of the 42, and 321 of the 642 ordinates have $\mu > 0$; per-ordinate spectra are written per direction grid to `toa_mean_radiance.csv` and `toa_mean_radiance_642.csv`).

Six comparison pairs (B − A): SC↔chain per δ, δ-on↔δ-off per method, octree↔unigrid-chain per δ. Plus the **accuracy ranking**: every run against `octree_ref_dM0` (`ranking.csv`).

---

4. Results

4.1 Convergence and efficiency

run	transport	δ-M	N_dir	n_iter	converged	wall [s]	⟨J⟩	mean F_z @TOA
uni_chain_dM0	chain	off	42	32	✓	1.6	0.5188	−2.5215
uni_chain_dM1	chain	M=1	42	**12**	✓	0.7	0.5188	−2.5214
uni_sc_dM0	sc	off	42	51	✓	16.6	0.4632	−2.5935
uni_sc_dM1	sc	M=1	42	**12**	✓	4.1	0.4633	−2.5933
octree_dM0	octree march	off	42	30	✓	37.9	0.4014	−2.5312
octree_dM1	octree march	M=1	42	**12**	✓	20.7	0.4016	−2.5308
uni_chain48_dM1	chain 48³	M=1	42	13	✓	5.4	0.5194	−2.5223
uni_sc48_dM1	sc 48³	M=1	42	12	✓	15.9	0.4640	−2.5927
octree_d4_dM1	octree march d4	M=1	42	12	✓	33.0	0.4114	−2.5319
octree_ref_dM1	octree march d5	M=1	642	34	✓	3393.7	0.3778	−2.5073
octree_ref_dM0	octree march d5	off	642	23	✓	2367.4	0.3778	−2.5074

- **delta-M is a large accelerator at 42 ordinates**: 32 → 12 (chain), 51 → 12 (SC), 30 → 12 (octree d3) iterations; wall time drops by ~2× everywhere and by ~4× for SC.
- **but not at 642 ordinates**: on the reference grid δ-M *increases* the iteration count (23 off → 34 on) and hence the wall time (2367 → 3394 s). The acceleration benefit is angular-grid dependent — at 42 ordinates the row-normalized HG operator at $g \approx 0.95$ is extremely forward-peaked and removing its g^1 moment pays; at 642 ordinates it no longer does.
- The unigrid TOA mean F_z reproduces the MANUAL §11 numbers exactly (−2.5215 unigrid, −2.5312 octree d3).
- **the 42 → 642 ordinate step moves the reference itself**: ⟨J⟩ 0.3840 → 0.3778 (−1.6%) and mean F_z @TOA −2.5320 → −2.5074 (+1.0%) — the 42-ordinate angular error inflates in-domain J and makes the net TOA flux too negative even on the well-resolved octree.
- **δ-M equivalence at the finest grid**: `octree_ref_dM1` vs `octree_ref_dM0` differ by 0.19% max / 0.005% mean in J — half the depth-3 gap (1.3%), confirming the $O(\tau_{cell})$ scaling of the explicit term and that the reference is δ-M-independent.

4.2 Mean intensity J

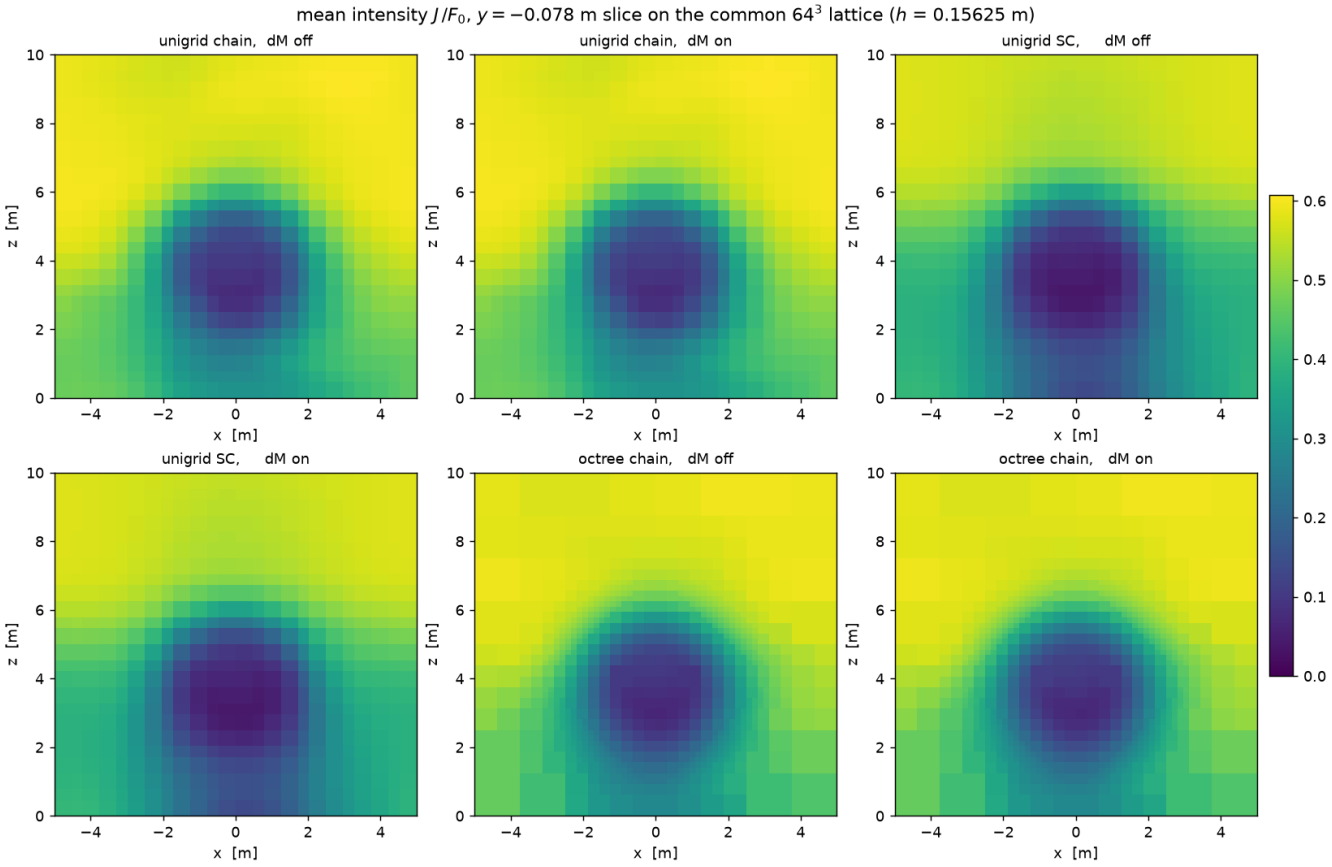


Figure 1: J slices

Band-mean J profiles (8 base-cell bands, `profiles.csv`; δ-on/off pairs agree to ≤3e−4, δ-off values shown):

z [m]	chain 24³	SC 24³	octree d3	chain 48³	SC 48³	octree d4	reference (642 ord.)
0.625	0.4372	0.3680	0.3095	0.4382	0.3660	0.3102	0.3388
3.125	0.4529	0.3629	0.3431	0.4539	0.3636	0.3587	0.3096
4.375	0.4964	0.4077	0.4154	0.4974	0.4105	0.4143	0.3186
6.875	0.5870	0.5631	0.5377	0.5872	0.5642	0.5443	0.5517
9.375	0.5893	0.5712	0.5850	0.5894	0.5709	0.5849	0.5840

Note the reference column: in the cloud core (z = 3.1–4.4 m) the reference J is far *lower* (0.31–0.32) than every depth-3-or-coarser run (0.41–0.50) — the coarse grids overestimate in-cloud radiation trapping; only the depth-4/5 octree resolution comes close. Compared with the 42-ordinate reference of the previous study version, the 642-ordinate reference lowers the cloud-core bands further (0.319/0.330 → 0.310/0.319) and raises the bottom band (0.319 → 0.339): part of the old octree-d3↔reference agreement in the bottom band was angular-error cancellation.

- **chain vs. SC on identical cells is the largest pairwise discrepancy of the six-run study** (max 31%, mean 9%): SC's extra numerical diffusion lets more radiation penetrate the optically thick core, where it is absorbed (ω = 0.9, τ = 10) — J is lower everywhere below and inside the cloud (0.368 vs 0.437 in the bottom band).
- **unigrid-chain vs. octree** (18% max, 2.8% mean) reproduces the anchor-5 pattern of the manual: the unigrid grazing-ordinate stencil error inflates J below the cloud (0.437 vs 0.310 in

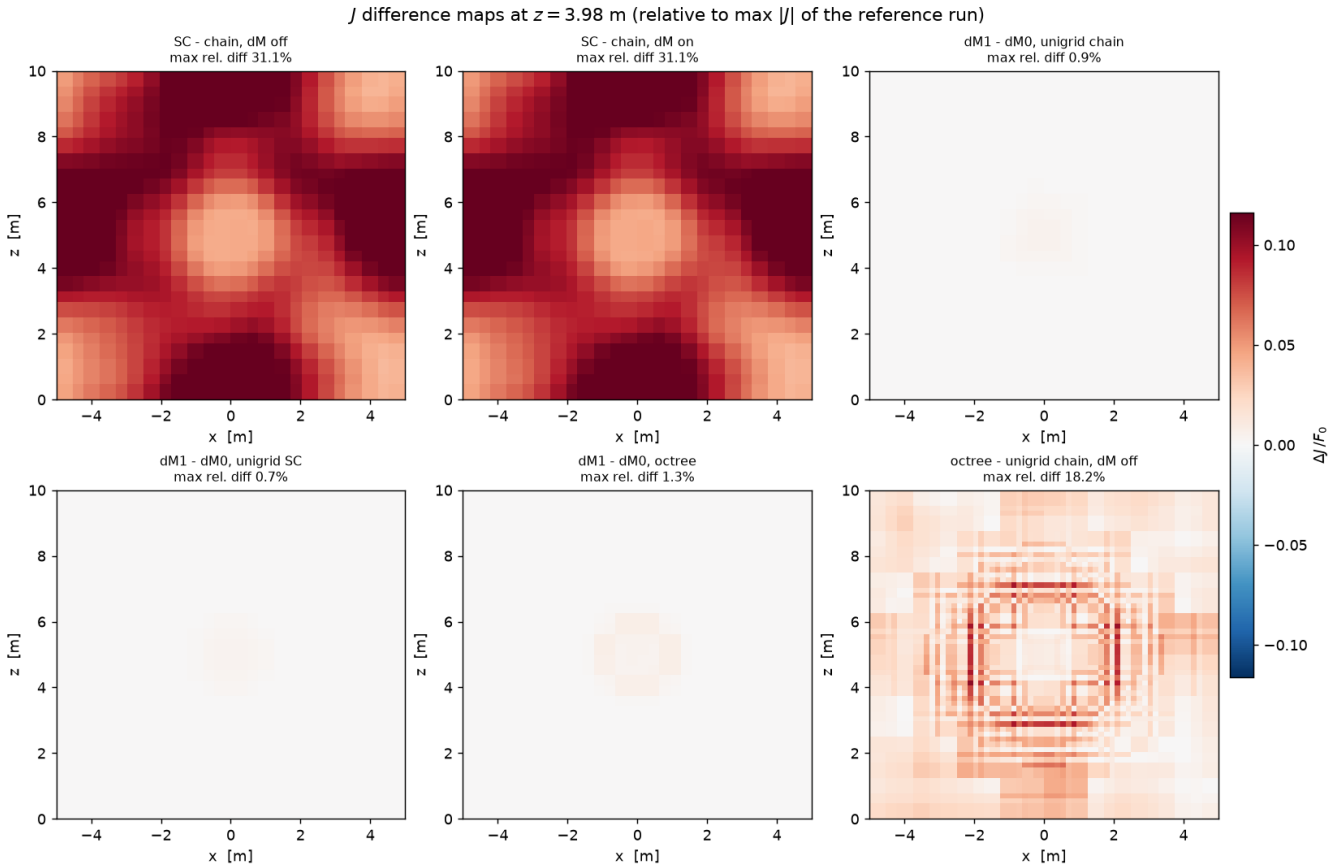
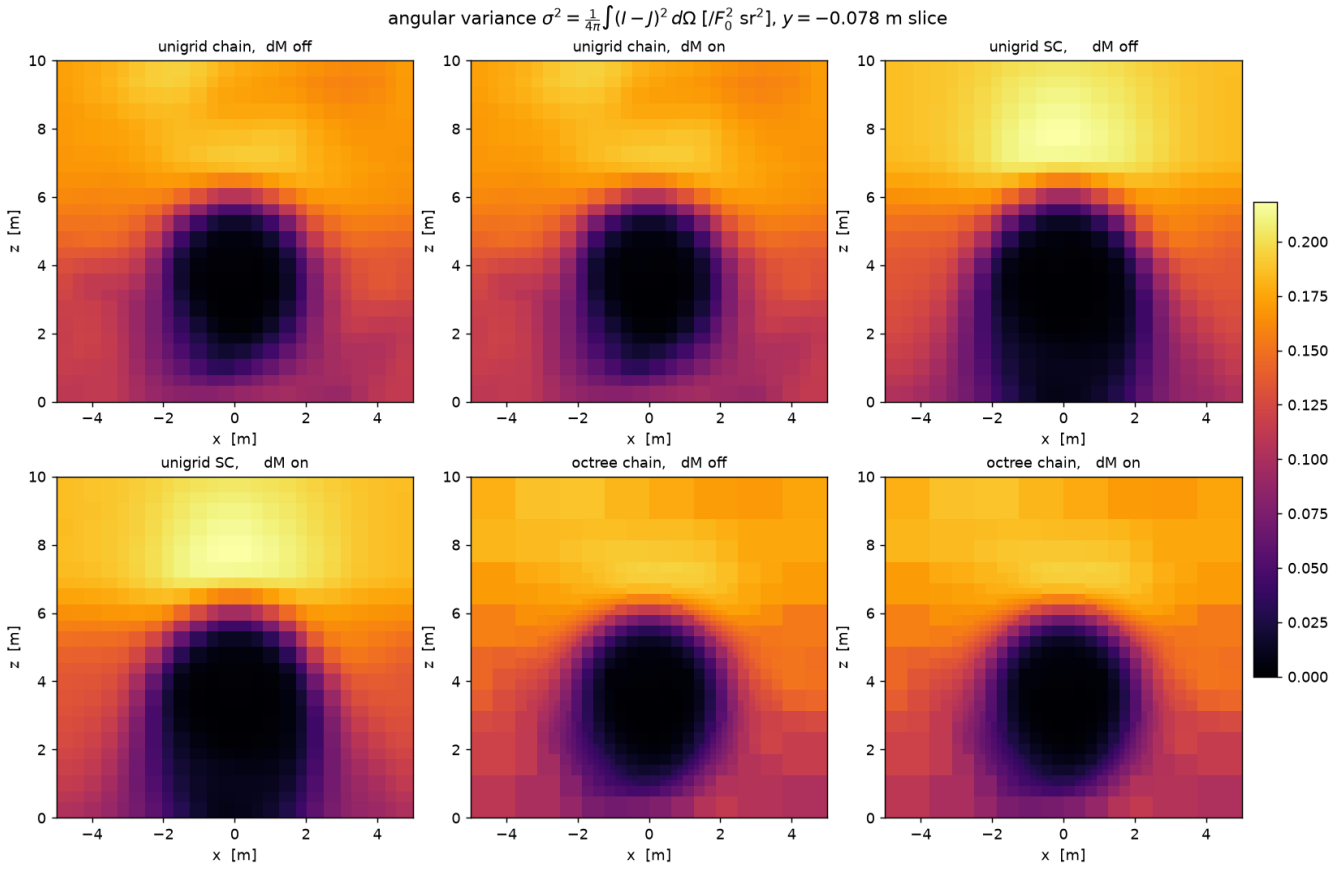


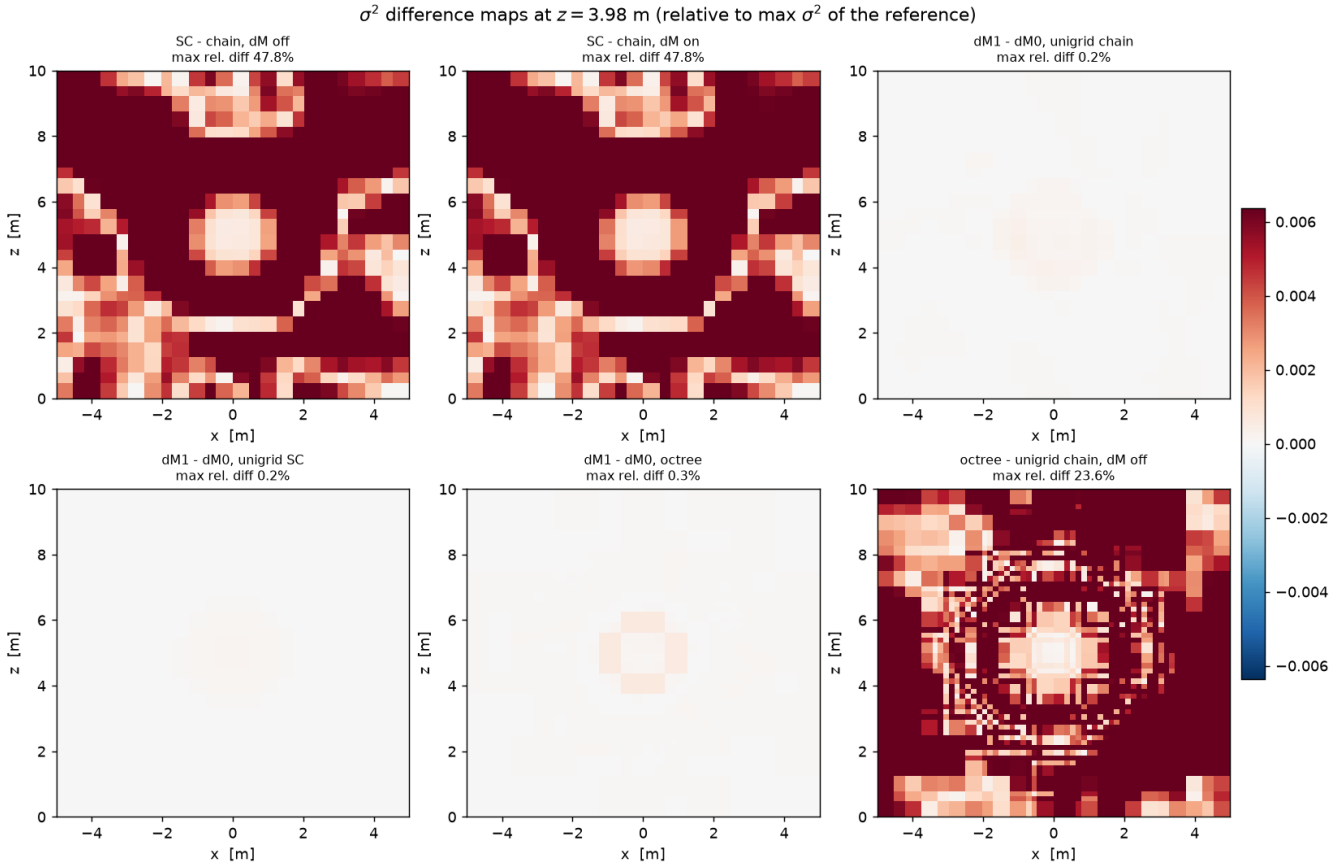
Figure 2: J difference maps

the bottom band, the manual’s 29%), with a ring structure around the cloud in the difference maps.

- **delta-M pairs are flat** (≤1.3% max, ~0.01–0.02% mean): the converged solution is independent of δ-M, as designed; the residual pointwise gap is the documented O(τ_cell) quadrature error of the explicit −ω g^M I term.

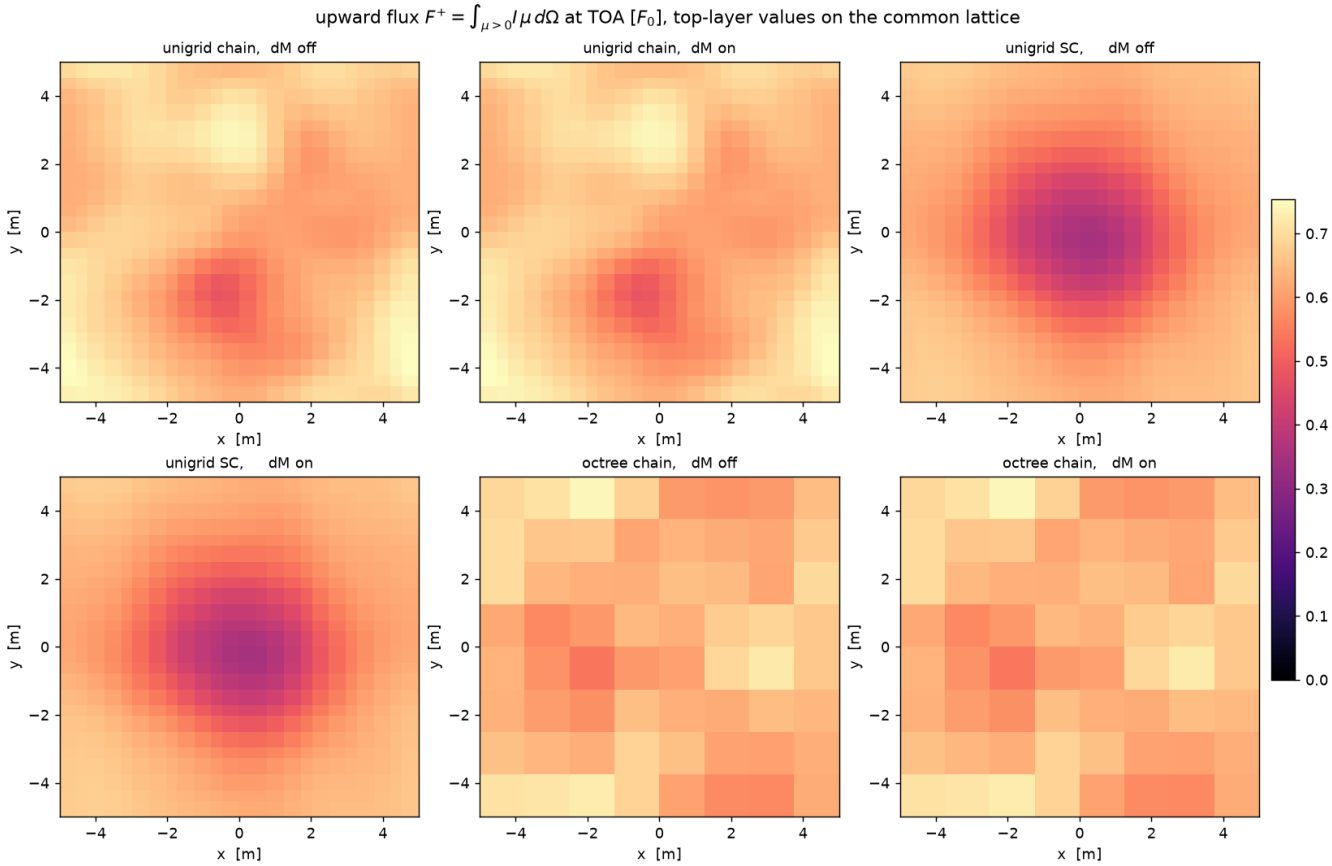
4.3 Angular variance

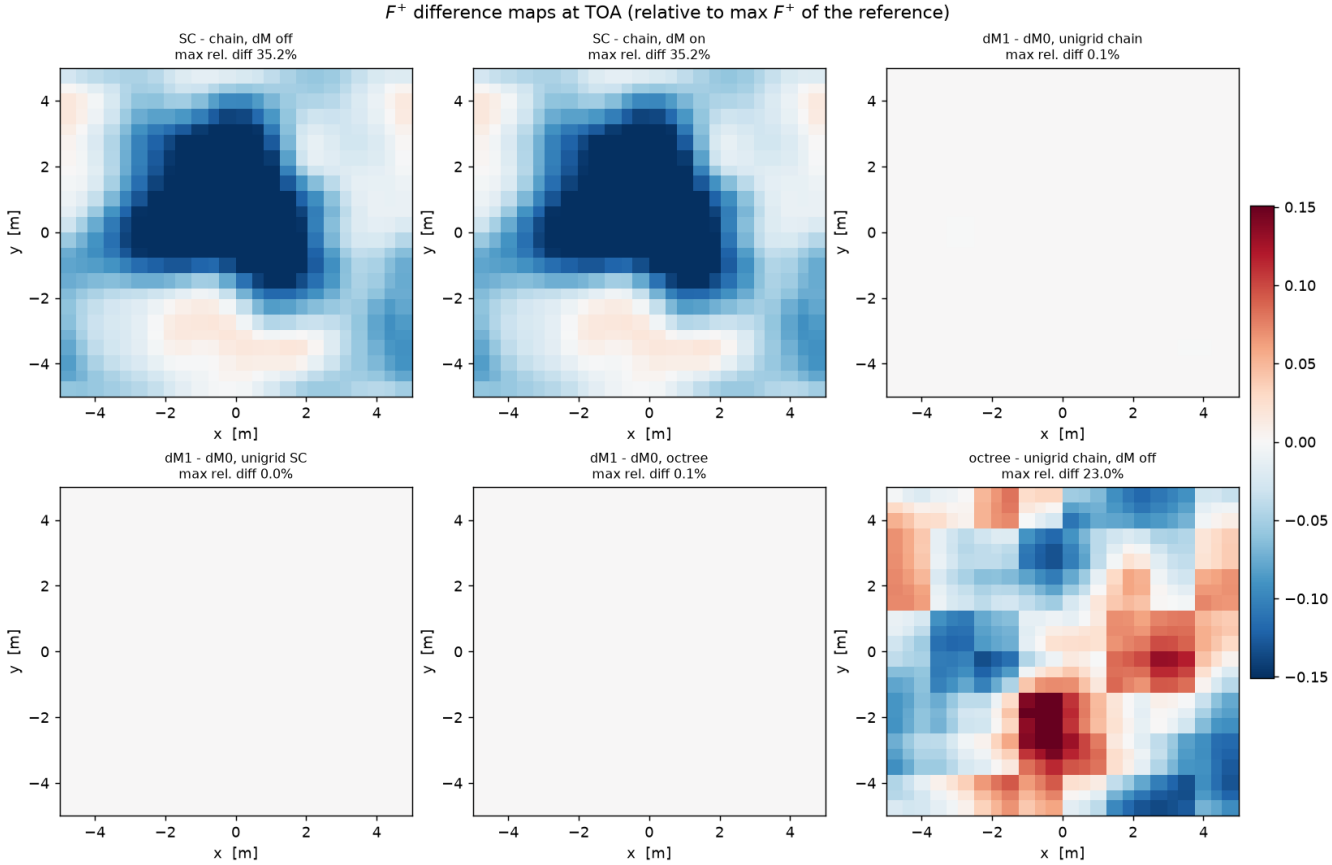


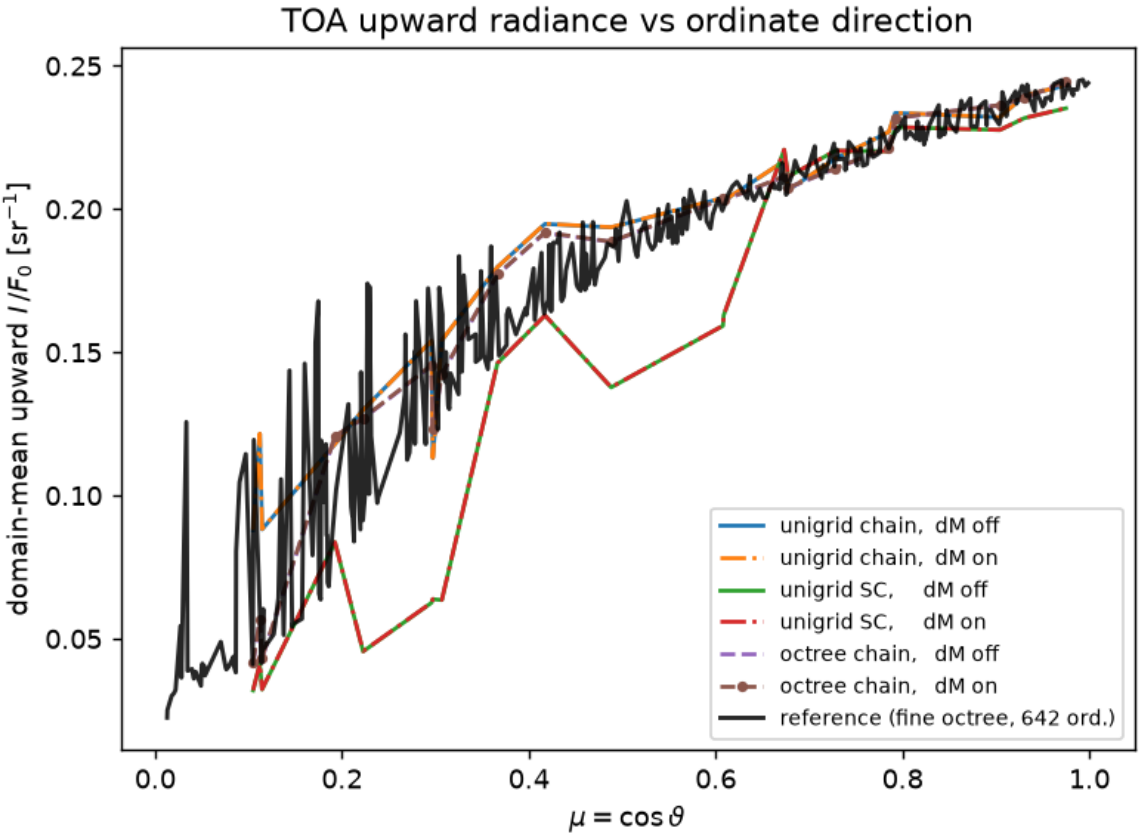


σ^2 is high in the vacuum (collimated beam + background = strongly anisotropic field) and collapses toward zero inside the cloud core where multiple scattering isotropises the field — the “hole” in all six panels. Scheme differences show up mostly in the *shadow region below the cloud* and in the beam path: SC’s diffusion smooths the beam shadow (47.8% max variance difference vs. chain), while the octree resolves the shadow boundary at 0.156 m. delta-M changes σ^2 by ≤0.3%.

4.4 Upward radiance at TOA







- The **chain** runs show a *bright* spot above the cloud (the scattering layer seen from above), the **SC** runs a deep *shadow* blob — the TOA-integrated signature of SC's extra diffusion (max F^+ difference 35%). The octree map is coarse by construction (top-layer cells are 1.25–2.5 m, the cloud is far below and the refinement is κ_e-driven).
- Per-ordinate domain-mean radiance (`toa_mean_radiance.csv`): chain and octree agree closely at all μ; SC collapses at grazing ordinates (μ = 0.10: 0.032 vs 0.088 chain / 0.041 octree) — the SC wrap-ring / grazing-path error — and lies between chain and octree at $\mu \gtrsim 0.7$.
- Domain-mean upward radiance: chain 0.179, octree 0.170, SC 0.142, reference (642 ord.) 0.168 [F_0 sr^{-1}]; consistently, the TOA net flux is least upward-escaping for SC (mean F_z −2.5935 vs −2.5215/−2.5312, reference −2.5074).

4.5 Pair statistics (from `pair_stats.csv`)

Relative differences of B − A, normalised by max|A| of the reference run (max over the common lattice / mean over the domain):

pair	J max	J mean	σ² max	σ² mean	F^+ max	F^+ mean
SC − chain, δ off	31.1%	9.1%	47.8%	6.2%	35.2%	9.8%
SC − chain, δ on	31.1%	9.1%	47.8%	6.2%	35.2%	9.8%
δ1 − δ0, unigrid chain	0.9%	0.01%	0.2%	0.02%	0.1%	0.02%
δ1 − δ0, unigrid SC	0.7%	0.01%	0.2%	0.01%	0.0%	0.02%

pair	J max	J mean	σ² max	σ² mean	F⁺ max	F⁺ mean
δ1 − δ0, octree	1.3%	0.02%	0.3%	0.04%	0.1%	0.05%
octree − unigrid chain, δ off	18.2%	2.8%	23.6%	3.2%	23.0%	7.6%

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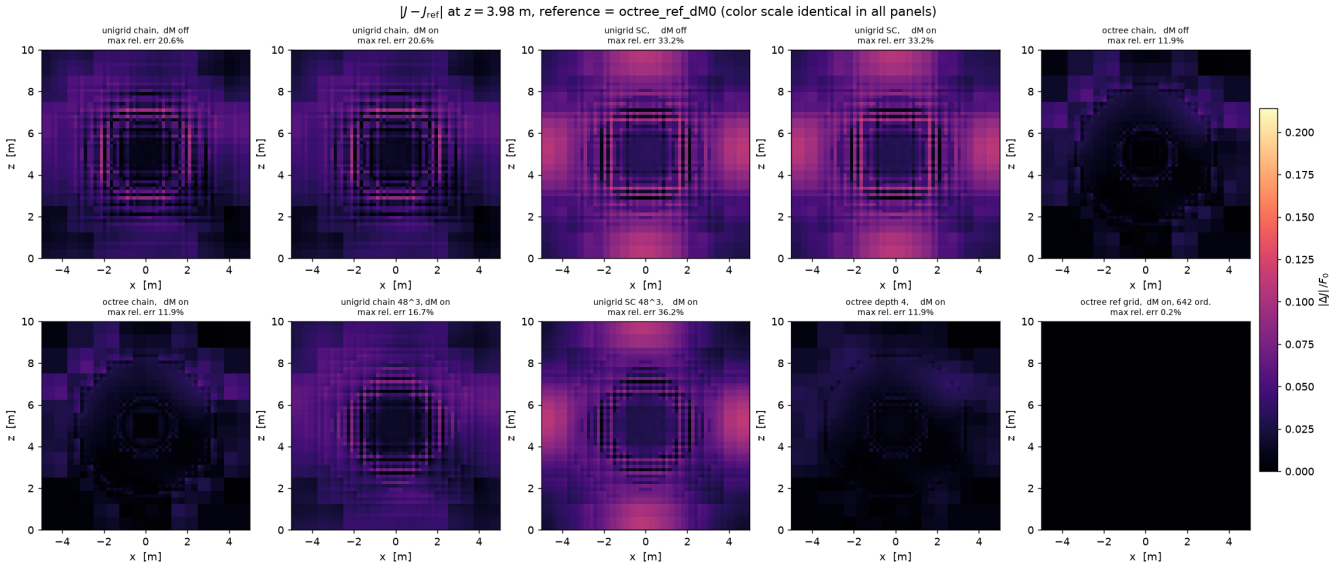
5. Which run is best? — accuracy ranking vs. the reference

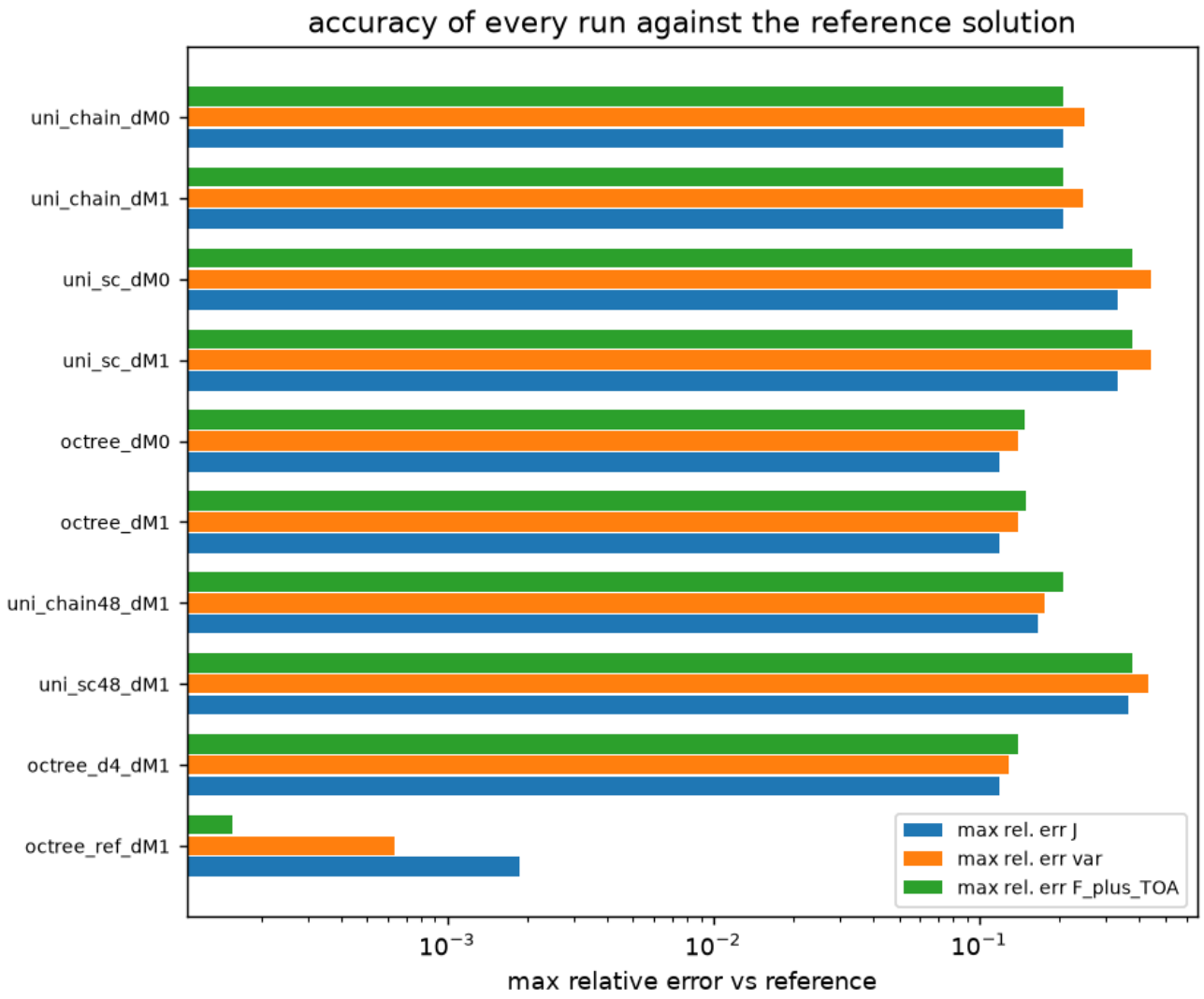
All pairwise numbers above compare approximations *to each other*. Ranking needs a solution that is demonstrably closer to the continuum limit: the fine-octree, **642-ordinate** reference `octree_ref_dM0` (section 2.6). Errors below are relative to max|·| of the reference field on the common lattice (`ranking.csv`, `err_bars.png`, `j_err_maps.png`). Because the reference resolves angle 15× better than the development grid, the errors of the 42-ordinate runs below are *space* + *angle* errors together — which is exactly what one wants when asking how far a practical run is from the truth.

5.1 Ranking

run	wall [s]	J err max	J err mean	σ² err max	F⁺ err max	ΔF_z @TOA
`octree_ref_dM0`	3393.7	0.2%	**0.005%**	0.1%	0.0%	+0.0001
`octree_d4_dM1`	33.0	11.9%	**1.85%**	12.9%	14.1%	−0.0245
`octree_dM1`	20.7	11.9%	**1.92%**	14.1%	14.9%	−0.0235
`octree_dM0`	37.9	11.9%	1.91%	14.0%	14.9%	−0.0238
`uni_chain48_dM1`	54.1	16.7%	**3.46%**	17.7%	20.7%	−0.0150
`uni_chain_dM1`	10.7	20.6%	**3.47%**	24.7%	20.6%	−0.0140
`uni_chain_dM0`	21.6	20.6%	3.46%	24.7%	20.6%	−0.0142
`uni_sc48_dM1`	115.9	36.2%	**6.18%**	43.4%	37.6%	−0.0853
`uni_sc_dM1`	4.1	33.2%	**6.29%**	44.2%	37.5%	−0.0860
`uni_sc_dM0`	16.6	33.2%	6.30%	44.2%	37.6%	−0.0861

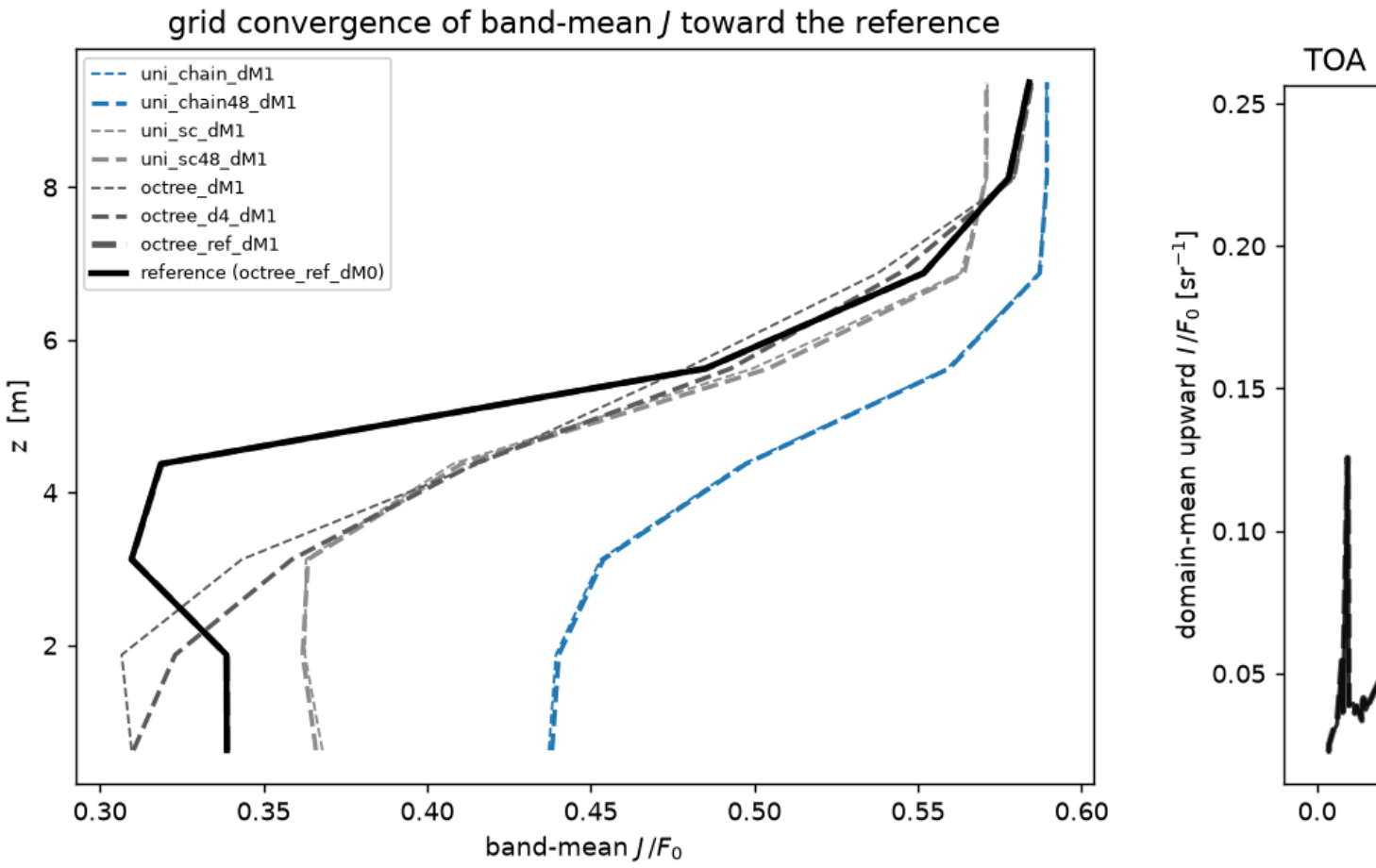
(reference `octree_ref_dM0` = 0 by construction; ΔF_z relative to −2.5074: chain −0.6%, SC −3.4%, octree d3 −0.9%; Δ⟨J⟩: chain +0.141, SC +0.086, octree d3 +0.024)





Accuracy order: octree > unigrid chain > unigrid SC (mean J error 1.9% vs 3.5% vs 6.3%). The gaps are much *narrower* than against the 42-ordinate reference of the previous study version (0.5% vs 2.7% vs 6.9%): with the angular error no longer cancelling, all 42-ordinate runs sit near a common **angular error floor**, and the octree's spatial advantage (which is real — see 5.2) is partly masked by it.

5.2 Convergence evidence (why the ranking is meaningful)



- **Octree** d3 → d4 (both 42 ordinates): mean J error 1.91% → 1.85%, max 11.9% → 11.9% — *flat*, in stark contrast to the 0.47% → 0.22% halving measured against the 42-ordinate reference. Against the 642-ordinate reference the octree's 42-ordinate **angular** error (~1.9% mean, ~12% max) dominates its remaining cell-size error.
- **Unigrid chain** 24^3 → 48^3 (8× cells): mean error 3.46% → 3.46%, max 20.6% → 16.7% — the mean is pinned at the angular floor; the remaining error is dominated by the **grazing-ordinate stencil direction error** (an angular error that does not shrink with h at fixed 42 ordinates) plus the common quadrature error.
- **Unigrid SC** 24^3 → 48^3: mean 6.30% → 6.18%, max 33.2% → 36.2% — likewise dominated by the angular grazing-ordinate wrap-ring error (section 4.4), not by cell size.
- **The reference's own angular error**: moving the reference grid from 42 to 642 ordinates changed ⟨J⟩ by 1.6% and TOA F_z by 1.0% — so the 42-ordinate quadrature error is a ~1–2% effect on scalars even for the octree, and the 642-ordinate reference's remaining angular error is correspondingly smaller again.

Putting the two rankings together: the octree's *cell-size* error really does halve per refinement level (0.47% → 0.22% → 0 when the angle is held at 42 ordinates), but at 42 ordinates every scheme additionally carries a scheme-dependent *angular* error (octree ~1.9%, chain ~3.5%, SC ~6.3% mean) that only N_dir removes. The unigrid schemes' angular error is the grazing-ordinate stencil/wrap error of section 4.4 — much larger than the octree's pure quadrature error, which is why the octree still wins the ranking and why refining the unigrid alone does not close the gap.

5.3 Verdict

- **Most accurate** (of the original six): `octree_dM1` — 1.9% mean J error against the 642-ordinate reference (0.47% against the 42-ordinate one), 20.7 s. The fine reference grid itself (`octree_ref_dM0`, 2367 s) is the right choice when the answer itself is the goal; its δ-M twin reproduces it to 0.005% mean but is *slower* (3394 s) because δ-M no longer accelerates at 642 ordinates.
- **Best accuracy per cost**: `octree_dM1` for physically trustworthy fields; `uni_chain_dM1` (0.7 s, 3.5% mean error) is the cheap workhorse when 4% is acceptable — but its error is angular and will not shrink with grid refinement at 42 ordinates.
- **delta-M on dominates off at 42 ordinates** (same converged fields, ≤1.3% max, 2–4× less wall time), but **not at 642** (34 vs 23 iterations on the reference grid). At cloud-core g ≈ 0.95 with the development grid it should be the default; on finely-resolved angular grids it is at best neutral.
- **SC is dominated on both axes**: less accurate *and* more expensive than chain at identical grids (24^3: 6.3% @ 4.1 s vs 3.5% @ 0.7 s).

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6. Findings

1. **delta-M (M = 1) does what it should at 42 ordinates**: it cuts the iteration count to 12 for every scheme (from 32/51/30), and the converged fields agree with the δ-off runs to ≤1.3% max / ≤0.05% mean — the residual is the known O(τ_cell) explicit-term quadrature gap (it halves to 0.19% at the finest grid), not a physics change. At the cloud-core g ≈ 0.95 with the development grid it should be the default setting. **New**: at 642 ordinates the acceleration disappears (23 → 34 iterations on the reference grid) — the benefit is tied to the coarse, strongly row-normalized 42-ordinate HG operator.
2. **Against the 642-ordinate reference, the accuracy order is octree > chain > SC** (mean J error 1.9% vs 3.5% vs 6.3%) — but all three sit near a common *angular error floor* of the 42-ordinate development grid: refining the octree d3 → d4 or the unigrid 24^3 → 48^3 barely moves the mean error any more. The octree's *cell-size* error still converges normally (0.47% → 0.22% → 0 against the 42-ordinate reference); what the new ranking exposes is that at 42 ordinates the angular part dominates for every scheme — smallest for the octree (pure quadrature error), much larger for the unigrid schemes (grazing-ordinate stencil/wrap errors).
3. **SC is systematically more diffusive**: lower J inside and below the cloud, deeper TOA shadow, collapsed grazing-ordinate radiance, and a ~3.4% too-small net upward flux at TOA. It is also ~5× slower per iteration than chain on the unigrid (wrap-around ring passes).
4. **The octree buys both resolution and path fidelity**: finer cells where κ_e demands them (0.078 m in the reference cloud core) *and* marching along the true ordinate directions; its exact-march transport removes the chain's grazing-ordinate stencil error, which is why unigrid-chain vs. octree is the manual's anchor-5 pattern.
5. **δ-M is scheme- and grid-agnostic in the converged solution**: the δ-on↔δ-off differences are ~2 orders of magnitude smaller than any scheme comparison, so the comparison is not confounded by the scattering-operator split (its *iteration* benefit is not angular-grid-agnostic — see 1).
6. **Angular error is now measured, not assumed**: the 42 → 642 ordinate step on the reference grid moves ⟨J⟩ by 1.6% and TOA F_z by 1.0%, and the ranking gaps of the 42-ordinate runs are dominated by their angular error. The full direction-convergence study (42/162/642 for all schemes, phase 6) should re-run this ranking to complete the picture.

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7. Interactive 3D scenes

Self-contained vtk.js HTML (open directly in a browser — orbit, actor visibility toggling, colour maps; no server, no Python needed). One scene per run (κ_e isosurfaces at 0.5/1/2/3 m^{-1} + J and σ^2 cut planes

at x = 0, y = 0, z = 4) plus difference scenes (|ΔJ| isosurfaces + slices), all regenerated 2026-10-15 on macOS:

```
plot/compare_transport/scene_uni_chain_dM0.html   scene_uni_chain_dM1.html
plot/compare_transport/scene_uni_sc_dM0.html      scene_uni_sc_dM1.html
plot/compare_transport/scene_octree_dM0.html      scene_octree_dM1.html
plot/compare_transport/scene_octree_ref_dM0.html
plot/compare_transport/scene_diff_chain_vs_sc_dM0.html
plot/compare_transport/scene_diff_uni_vs_octree_dM0.html
plot/compare_transport/scene_diff_ref_vs_chain24_dM0.html
plot/compare_transport/scene_diff_ref_vs_sc24_dM0.html
plot/compare_transport/scene_diff_ref_vs_octree3_dM0.html
```

The three `ref_vs_*` scenes are the ranking story made visible: where each coarse run gets J wrong against the reference (chain: below/around the cloud; SC: everywhere diffuse; octree d3: no longer just the cloud core — a broad few-percent pattern strongest in the anisotropic vacuum around the cloud, the angular error floor of the 42-ordinate ordinates).

(pyvista's vtk.js export cannot carry volume mappers, hence slices + isosurfaces; the static volume overview is `vol_j.png`.)

8. Output files

file	content
`summary.csv`	run table (grid, N_dir, iterations, wall, ⟨J⟩, TOA F_z), all 11 runs
`ranking.csv`	per-run error vs the reference (J, σ^2, F^+, scalars)
`profiles.csv`	band-mean J profiles, all 11 runs
`toa_mean_radiance.csv`	per-ordinate domain-mean upward radiance at TOA, 42-ordinate runs
`toa_mean_radiance_642.csv`	same for the two 642-ordinate reference runs
`pair_stats.csv`	max/mean relative differences of all six pairs × J, σ^2, F^+
`j_slices.png`, `j_profiles.png`, `j_diff.png`	mean-intensity figures
`j_err_maps.png`	|J − J_ref| at the cloud plane, every run vs reference
`j_convergence.png`	band-mean J: refinement sequences + reference
`err_bars.png`	max relative error vs reference, all runs (log scale)
`var_slices.png`, `var_diff.png`	angular-variance figures
`toa_fup_maps.png`, `toa_fup_diff.png`, `toa_mu_spectra.png`	TOA radiance figures
`toa_convergence.png`	TOA μ-spectra: refinement sequences + reference
`vol_j.png`	static volume renderings, base six runs + reference
`scene_*.html`	interactive 3D scenes (gitignored, regenerate by rerunning)
`_solve_cache/`	per-run solved fields (gitignored; reruns resume)